

Practice Problems on Combinational Circuits (Part 1)

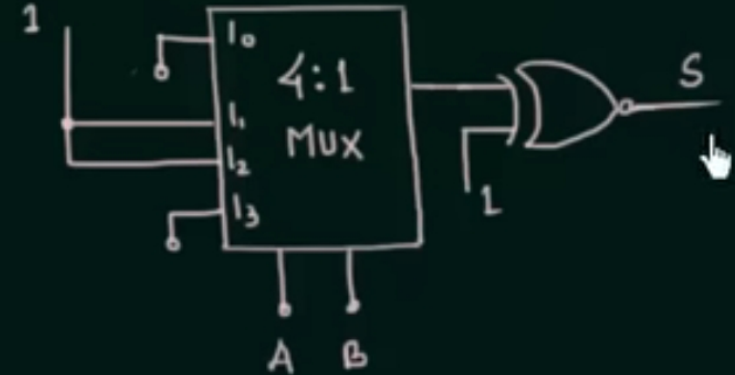
Question 1: The circuit shown does not represents

(a) $S(A,B) = \Sigma(1,2)$

(b) EXOR gate with
A and B as input

(c) $S(A,B) = \pi(0,3)$

(d) $A'B' + AB$



Solution: -



Practice Problems on Combinational Circuits (Part 1)

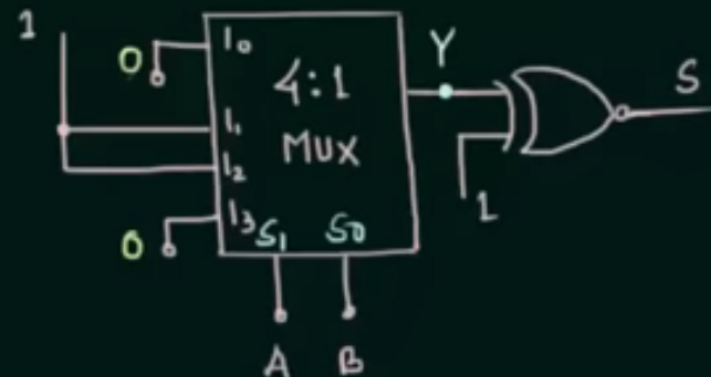
Question 1: The circuit shown does not represents

(a) $S(A,B) = \Sigma(1,2)$

(b) EXOR gate with
A and B as input

(c) $S(A,B) = \pi(0,3)$

(d) $A'B' + AB \equiv A \oplus B \neq S$



Solution: -

	S_1 A	S_0 B	
m_0	0	0	$I_0 = 0$
m_1	0	1	$I_1 = 1$
m_2	1	0	$I_2 = 0$
m_3	1	1	$I_3 = 0$

$Y = S = \Sigma m(1,2) = \Pi(0,3)$

$Y = \overline{A}\overline{B} \cancel{I_0} + \overline{A}B \cancel{I_1} + A\overline{B} \cancel{I_2} + AB \cancel{I_3}$

$Y = \overline{A}B + A\overline{B} = A \oplus B$

$$S = Y \oplus 1$$

$$S = Y'1' + Y.1$$

$$S = 0 + Y$$

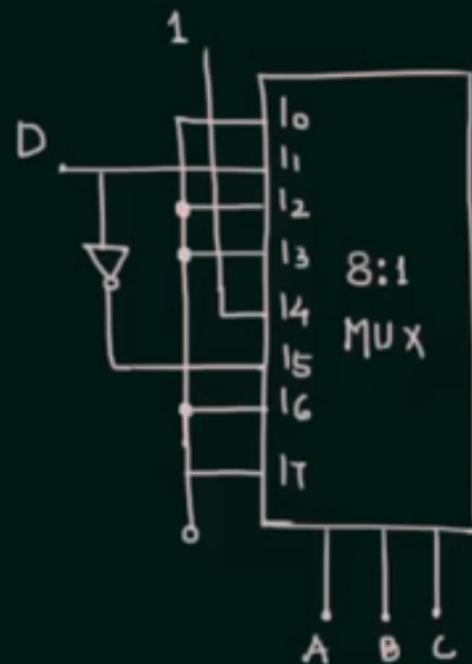
$$S = Y = A \oplus B = \overline{A}B + A\overline{B}$$



Question 2: The circuit below represents function $X(A,B,C,D)$ as:

- (a) $\Sigma(3,8,9,10)$ (b) $\Sigma(3,8,10,14)$
 (c) $\pi(0,1,2,4,5,6,7,11,12,13,15)$ (d) $\pi(0,1,2,4,5,6,7,10,12,13,15)$

Solution: -



Question 2: The circuit below represents function $X(A,B,C,D)$ as:

- $\Sigma(3,8,9,10)$

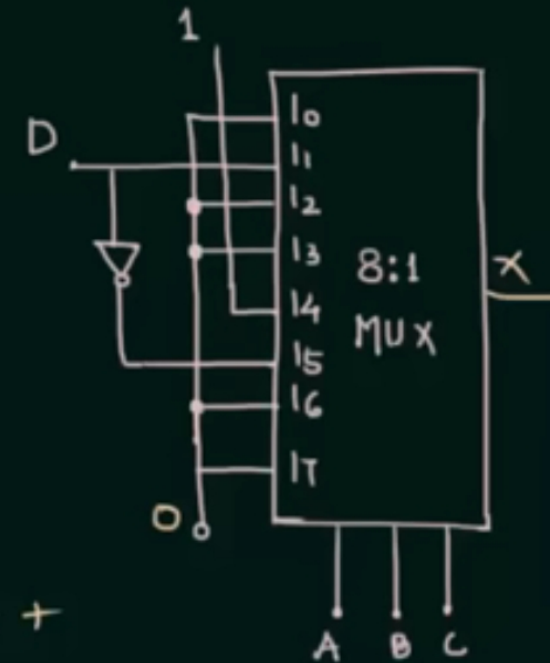
$\Sigma(3,8,10,14)$
- $\Sigma(0,1,2,4,5,6,7,11,12,13,15)$

$\pi(0,1,2,4,5,6,7,10,12,13,15)$

Solution: -

A	B	C	X
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

$$\begin{aligned}
 X &= \overline{A}\overline{B}\overline{C}I_0 + \overline{A}\overline{B}CI_1 + \overline{A}B\overline{C}I_2 + \\
 &\quad \overline{A}BCI_3 + A\overline{B}\overline{C}I_4 + A\overline{B}CI_5 + \\
 &\quad AB\overline{C}I_6 + ABCI_7 \\
 X &= \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}\overline{C} + A\overline{B}\overline{C}\overline{D} + \\
 &= \Sigma m(3, 9, 8, 10)
 \end{aligned}$$



$$\begin{array}{r}
 A\overline{B}\overline{C}(D + \overline{D}) \\
 \hline
 A\overline{B}\overline{C}D + A\overline{B}\overline{C}\overline{D} \\
 \hline
 1001 \quad 1000
 \end{array}$$



Practice Problems on Combinational Circuits (Part 2)



Question 1: Consider the following statements:

A multiplexer

1. selects one of the several inputs and transmits it to a single output.
2. routes the data from a single input to one of the many output.
3. converts parallel data into serial data.
4. is a combinational circuit.

which of these statements are correct?

(a) 1,2 and 4

(b) 2,3 and 4

(c) 1,3 and 4

(d) 1,2 and 3

Solution:



Practice Problems on Combinational Circuits (Part 2)

Question 1: Consider the following statements:

IES - 2000

A multiplexer

- ✓ 1. selects one of the several inputs and transmits it to a single output.
- ✗ 2. routes the data from a single input to one of the many output.
- ✓ 3. converts parallel data into serial data.
- ✓ 4. is a combinational circuit.

which of these statements are correct?

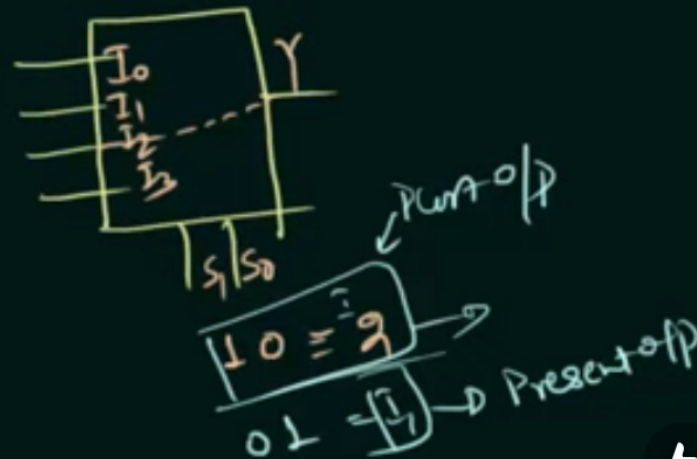
✗ (a) 1, 2 and 4

✗ (b) 2, 3 and 4

✗ (c) 1, 3 and 4

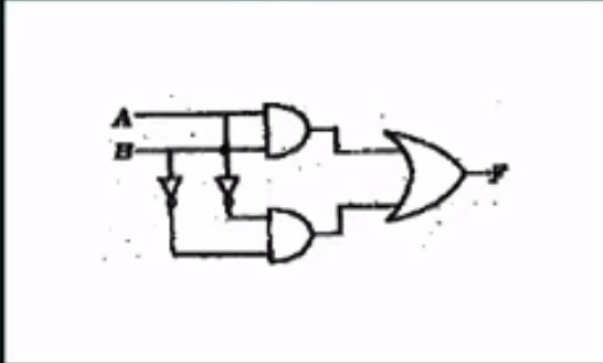
✗ (d) 1, 2 and 3

Solution:

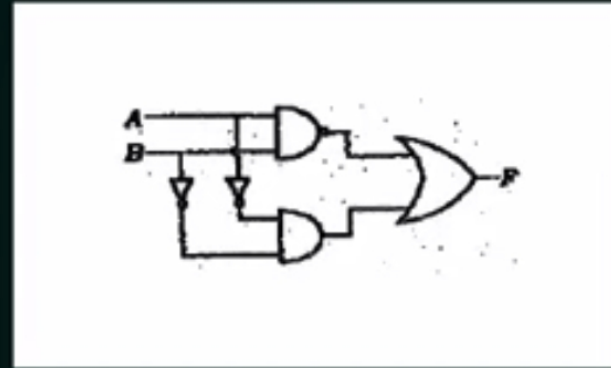


Question 2: Which of the following figures represents the coincidence logic?

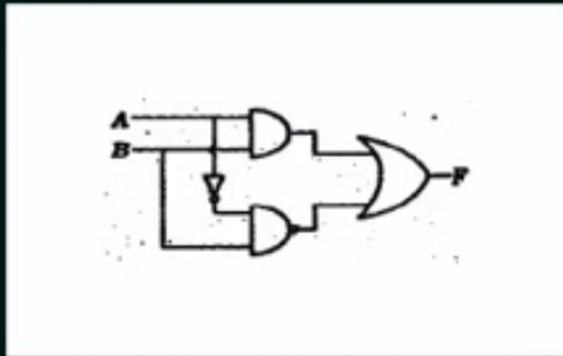
(a)



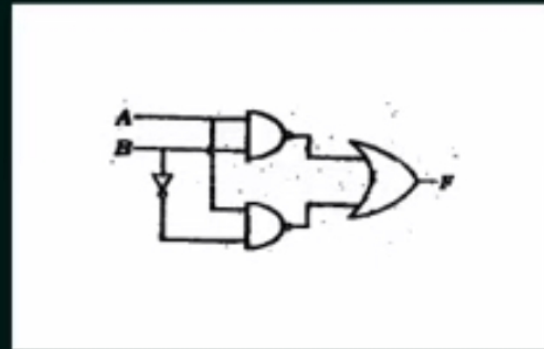
(b)



(c)

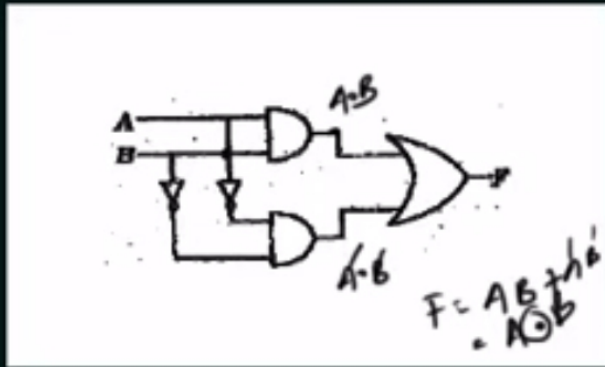


(d)

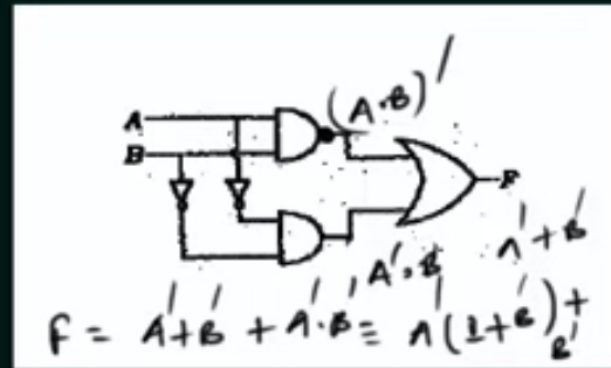


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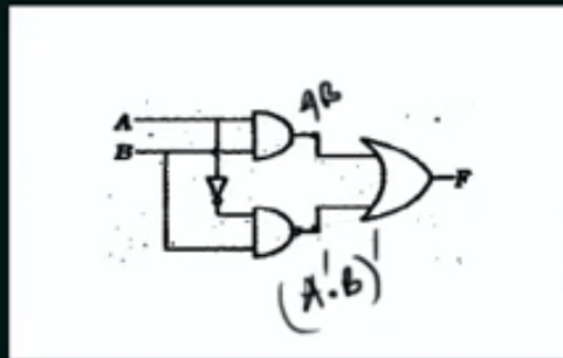
(a)



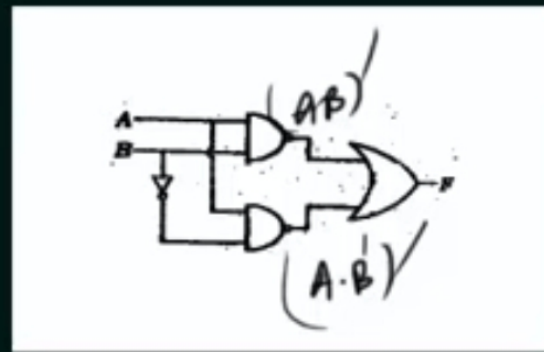
(b)



(c)



(d)



0 1	Circuit	Y = 1
0 1		
0 1		

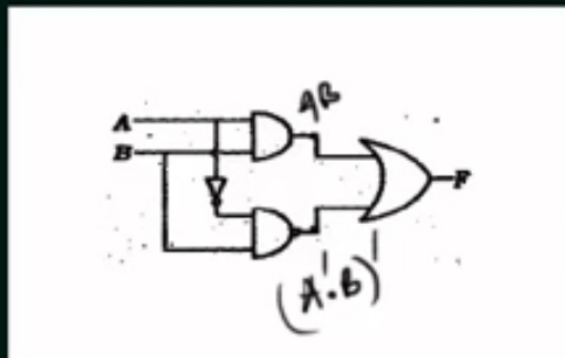
input ≥ 2
 $\Rightarrow$ of the same time
 $\Rightarrow$ output = 1

XNOR gate

A	B	F
0	0	1
0	1	0
1	0	0
1	1	1



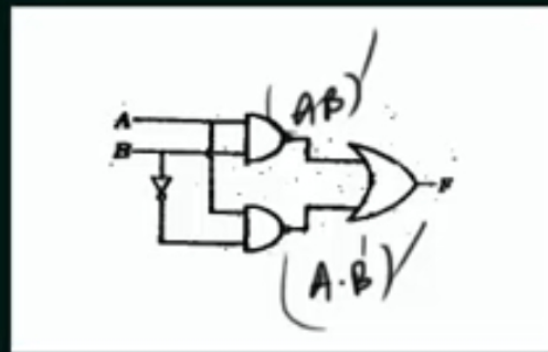
(c)



$$F = AB + A + B'$$

$$f = A + B'$$

(d)



$$F = A' + B' + A' + B$$

$$= A' + 1$$

$$f = 1$$

XNOR gate

A	B	F
0	0	1
0	1	0
1	0	0
1	1	1

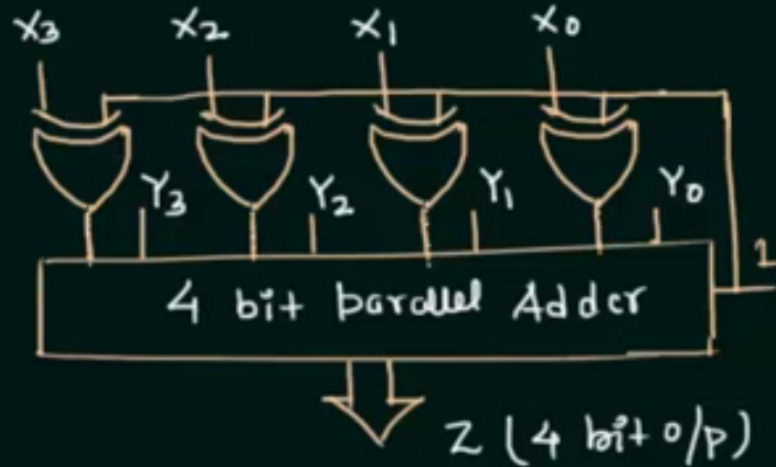


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Practice Problems on Combinational Circuits (Part 3)

Question: 1 Identify the correct statement with respect to following circuit.



a) it outputs $X+Y$

b) it outputs $Y-X$

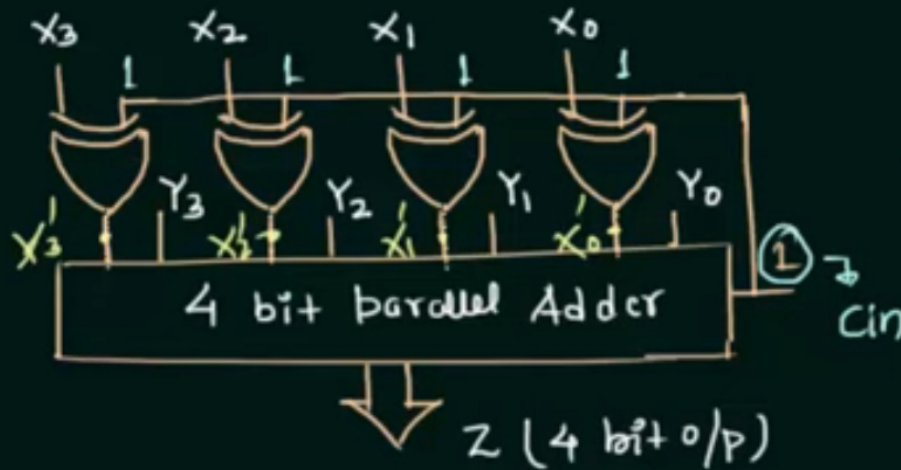
c) it outputs $X+1$

d) it outputs $Y+1$



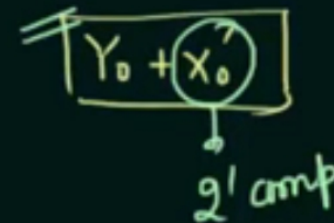
Practice Problems on Combinational Circuits (Part 3)

Question: 1 Identify the correct statement with respect to following circuit.



4-bit Adder/Sub.

$$\begin{aligned}
 A \oplus B &= A'B + B'A \\
 &= A'1' + 1 \cdot A' \\
 &= B + A' \\
 &= A'
 \end{aligned}$$



$$\begin{array}{ll}
 Y_0 - X_0 & Y_2 - X_2 \\
 Y_1 - X_1 & Y_3 - X_3 \\
 \text{Adding 2's comp} & \equiv \text{sub}
 \end{array}$$

a) it outputs $X+Y$

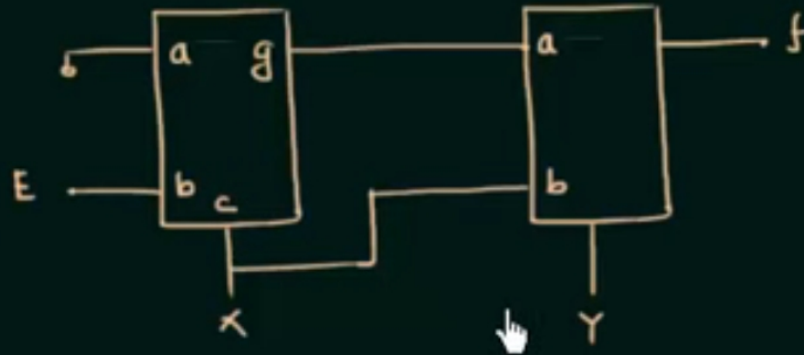
b) it outputs $Y-X$

c) it outputs $X+1$

d) it outputs $Y+1$



Question: 2 Consider the circuit shown below. The output of 2:1 is given by the function $g = ac + bc'$



Then f is:

(a) $X \text{ XOR } Y$

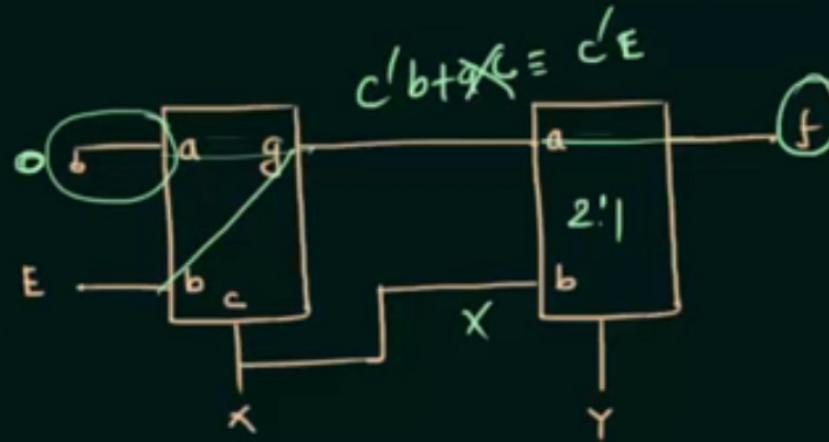
(b) $XY' + EX'Y$

(c) $XE + XY'$

(d) None of these



Question: 2 Consider the circuit shown below. The output of 2:1 is given by the function $g = ac + bc'$



Select Variable = c

$$\begin{array}{ll} c = 0 & \cancel{g=a} \quad g=b \\ c = 1 & \cancel{g=b} \quad g=a \end{array}$$

$$g = \frac{c'a + bc}{c'b + ac}$$

Then f is:

(a) $X \text{ XOR } Y$

(c) $XE + XY'$

~~(b) $XY' + EX'Y$~~

(d) None of these

$$f = Y'X + YX'E$$



Practice Problems on Combinational Circuits (Part 4)

Que: - Implement the following boolean expression using an 8:1 multiplexer
 $F(A,B,C,D) = \sum m(1,3,5,10,11,13,14) + d(0,2)$

Sol: -

AB \ CD				
	00	01	11	10
00				
01				
11				
10				



Practice Problems on Combinational Circuits (Part 4)

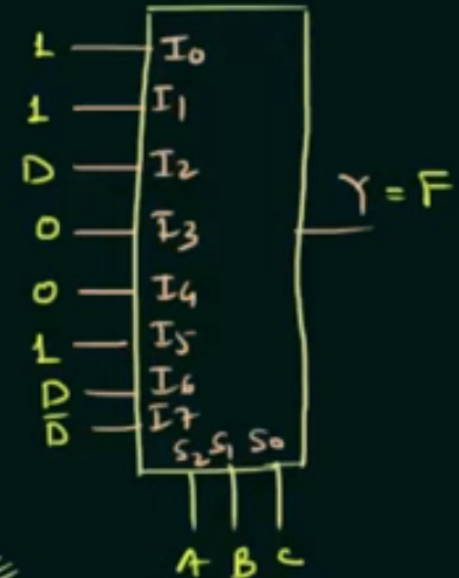
Que: - Implement the following boolean expression using an 8:1 multiplexer

$$F(A, B, C, D) = \sum m(1, 3, 5, 10, 11, 13, 14) + d(0, 2)$$

Sol: -

AB \ CD				
	00	01	11	10
00	X	1	1	X
01		1		
11		1		1
10			1	1

A	B	C	
s_2	s_1	s_0	Y
0	0	0	$I_0 = 1$
0	0	1	$I_1 = 1$
0	1	0	$I_2 = D$
0	1	1	$I_3 = 0$
1	0	0	$I_4 = 0$
1	0	1	$I_5 = 1$
1	1	0	$I_6 = D$
1	1	1	$I_7 = \bar{D}$

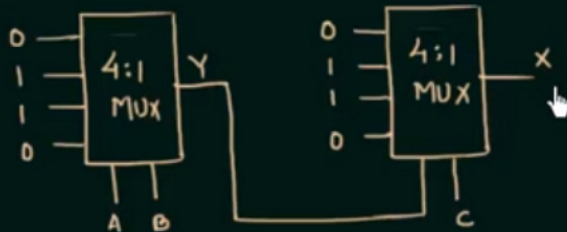


$$\begin{aligned} s_0 &= C \\ s_1 &= B \\ s_2 &= A \end{aligned}$$



[Home Work]

Question: In the following circuit, X is given by



(a) $X = AB'C' + A'BC' + A'B'C + ABC$

(b) $X = A'BC + AB'C + ABC' + A'B'C'$

(c) $X = AB + BC + AC$

(d) $X = A'B' + B'C' + A'C'$

Que: - Design a combinational circuit with three inputs, x, y , and z , and three outputs, A, B , and C .
When the binary input is 0,1,2, or 3, the binary output is two greater than the input. When the binary input is 4,5,6, or 7, the binary output is two less than the input.

Sol: -

